From Actionless Robot Demonstrations to Policy Data: A Lift Pilot of Inverse Dynamics, Visual Descriptions, and Hosted Decision-Based Selection

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Abstract

We study whether a pretrained vision model helps turn robot demonstrations whose actions are hidden from the learning pipeline into useful policy data. In a robomimic Lift pilot, 24 action-labeled demonstrations train local inverse-dynamics and grounding models; 16 further training demonstrations provide observations for pseudo-labeling. A policy trained on 24 gold demonstrations succeeds on 14.7% of closed-loop trials. Adding 16 local inverse-dynamics pseudo-labeled demonstrations yields 32.7%; adding 16 Gemini 3.6 Flash semantic descriptions grounded into actions yields 34.0%. The prespecified visual-versus-inverse comparison is +1.3 percentage points (original paired-start 95% interval $[-9.3, +11.3]$), which does not establish an advantage. Filtering eight pseudo-labeled demonstrations yields 42.0% for the inverse model and 46.7% for a local inverse-plus-visual hybrid; the latter difference is uncertain across three training seeds. In a later follow-up, hosted TypeSafe Jev 1.13 judged the *cached text descriptions* to select eight of the same hybrid pseudo-labeled demonstrations. This policy reached 38.7%, versus 42.0% when selection used Gemini's confidence and 46.7% with the existing hybrid risk score. Hosted Jev's held-out error-risk ranking was near chance (AUC 0.491). These are small, single-task pilots with no locked-test evaluation. They support the utility of pseudo-labeled data but do not establish added value from the visual or hosted decision layers over the corresponding local controls.

1 Question and scope

Action-conditioned robot training needs pairs of observations and executable controls. Videos and state trajectories often lack accessible action labels. The motivating Jev concept proposes an auditable data layer that infers useful transition descriptions or action distributions, then exports trajectories for downstream training. This study tests a limited part of that proposal: whether pseudo-actions inferred from one simulated manipulation task improve an independently trained behavior-cloning (BC) policy. No action-conditioned world model was trained or evaluated here. The raw robomimic files contain actions for all demonstrations; the experimental software masks the actions of the 16 pseudo-label targets. This is a controlled missing-label simulation, not a collection of naturally actionless videos.

The study has three distinct components. A trained inverse-dynamics model (IDM, arm B2) predicts native actions from frozen DINOv2 visual features. Gemini 3.6 Flash reads rendered RGB frames and returns a 24-dimensional semantic description; a trained grounding network maps that description and current features into native actions. An inverse-dynamics/semantic hybrid combines B2 predictions with the Gemini descriptions. TypeSafe Jev 1.13, accessed through OpenRouter, accepts text or JSON and returns typed decisions, not images or continuous robot actions (TypeSafe AI, 2026; OpenRouter, 2026). It was first used in the later E3b data-selection follow-up. Historical experiment IDs containing "JV" denote the Gemini-grounded or hybrid architecture; they do not indicate a TypeSafe Jev call.

2 Data, models, and evaluation

Data and controller. The Lift dataset has 200 episodes. A fixed split reserves 120 for training, 20 for development, 20 for calibration, and 40 for a locked test. The pilot uses the first 40 training episodes: 24 with visible actions for annotator and policy training, plus 16 whose action arrays are hidden from pseudo-label generation. Eight development episodes inform selection, eight calibration episodes fit risk maps, and 12 other development episodes form an evaluation holdout. The 40-episode test remains untouched. The 7-dimensional OSC_POSE controls are bounded in $[-1, 1]$, with a binary gripper channel; the recorded controller runs at 20 Hz. Grounding models receive only the 24 training action labels. The 40-gold policy reference also uses the native actions of the 16 otherwise masked training episodes and therefore has a larger label budget.

Pseudo-action generation. Local B2 predicts an action distribution from DINOv2 features of short observed transitions. Local B3 is a feature-difference reconstruction comparator; no B3 policy was retrained in E3. The Gemini semantic path sends four 256×256 agent-view images around each transition to `google/gemini-3.6-flash`, encodes its fixed JSON response as a 24-dimensional vector, and trains a small local grounding network on 24 gold episodes. A separate direct-output query asks Gemini for seven native-action values and supplies three fixed examples from the same label budget; this is a reconstruction diagnostic, not a policy arm. The hybrid grounding network takes B2 outputs, Gemini semantics, and visual features. Grounding, risk fitting, and policy training are distinct stages.

Risk and policy protocol. An action is marked bad when its mean position error exceeds the development-set B2 median threshold, its mean rotation error exceeds the corresponding threshold, or its gripper sign is wrong. The thresholds are 0.1168 and 0.0242 in native action units. Each method's uncertainty score is mapped to empirical error on the eight calibration episodes with isotonic regression. Filtering ranks the 16 unlabeled *episodes* by mean predicted risk and keeps eight; it does not discard individual transitions. Each policy has a 19-dimensional low-level observation input, a two-layer 256-unit BC network, and 3,000 optimizer updates. Three independently initialized policies per arm run on the same 50 simulator starts (seeds 10000–10049), for 150 closed-loop trials per arm and a 200-step horizon. The success metric is Lift's simulator success check. Policy seeds, not rollouts, are the limited training replication.

Later hosted decision experiment. E3b was specified *after* E3 outcomes were known. It sent existing Gemini semantic JSON, not images or native labels, to `typesafe/jev-1.13`. Three fixed Noul questions judged motion ambiguity, contact ambiguity, and internal inconsistency. The largest returned probability served as an uncalibrated uncertainty score; a new isotonic map was fitted on 383 calibration transitions. A matched control used $1-$Gemini confidence with the same calibration code. Each method kept eight of the same 16 episodes. Both E3b policies reused the *same saved hybrid pseudo-action arrays*, the same 24 gold episodes, policy settings, and paired evaluation starts. The original hybrid-risk filter is a historical comparator with a different existing calibration implementation. Exact repeated semantic inputs were cached, so 1,646 scored transitions required only 72 distinct hosted requests.

3 Closed-loop policy results

Table 1 and Fig. 1 report all evaluated policy arms. P0, B2, and the 40-gold reference anchor the interpretation. The unfiltered Gemini-grounded arm exceeded P0 by 19.3 percentage points, but only by 1.3 points over B2. The latter interval includes substantial benefit and harm; it is not an equivalence test. The unfiltered hybrid was below B2. The filtered Gemini-grounded arm also fell to 26.0%, a result that the original manuscript omitted from its overall conclusion.

Table 1: Lift policy success. Every row except the 40-gold reference uses 24 gold demonstrations. "All" means 16 pseudo-labeled demonstrations; "filtered" means eight. Each mean covers three policy seeds and 50 shared starts per seed. E3b rows reuse E3 hybrid pseudo-actions and vary only the episode selector.

Policy data and selector	Mean success
P0: 24 gold only	14.7%
Local B2: all 16 pseudo-labeled	32.7%
Local B2: filtered eight	42.0%
Gemini semantics + local grounding: all 16	34.0%
Gemini semantics + local grounding: filtered eight	26.0%
Local B2 + Gemini hybrid: all 16	24.0%
Local hybrid: existing risk filter	46.7%
Same saved hybrid actions: Gemini-confidence filter (E3b)	42.0%
Same saved hybrid actions: hosted TypeSafe Jev filter (E3b)	38.7%
P-REF: 40 gold demonstrations	45.3%

The E3 plan prespecified Gemini-grounded versus P0, Gemini-grounded versus B2, and unfiltered hybrid versus B2. Their reported pooled paired-start differences were +19.3 points (95% interval $[+9.3, +29.3]$), +1.3 ($[-9.3, +11.3]$), and −8.7 ($[-19.3, +2.0]$), respectively. These original intervals resample 150 paired outcomes, although each set of 50 shares one trained policy. A two-way seed/start resampling sensitivity interval for Gemini-grounded versus P0 spans approximately $[-5.3, +42.0]$ points; with only three seeds, its width cautions against a strong generalization from the favorable original pooled interval. Neither unfiltered visual comparison establishes superiority to B2.

The 46.7% filtered hybrid exceeds filtered B2's 42.0% by 4.7 points, but their per-seed differences are +20, −6, and 0 points. A paired two-way seed/start bootstrap interval is $[-14, +22]$ points. The contrast was not one of the E3 primary tests. Comparing filtered hybrid only with *unfiltered* B2 creates an avoidable selection mismatch; comparing 46.7% with the 40-gold reference's 45.3% does not establish superiority either. E3b's hosted-Jev filter is −3.3 points relative to the matched Gemini-confidence filter (95% two-way interval $[-17.3, +10.7]$) and −8.0 points relative to the existing hybrid filter ($[-22.7, +6.0]$). The two E3b comparisons were written down before E3b outputs, but after E3 results were available. No multiplicity adjustment was made; all intervals are descriptive.

4 Action and selection diagnostics

Figure 3 separates action reconstruction from policy utility. On the 12-episode evaluation holdout, local B2 and B3 have position-action mean absolute error (MAE) 0.1252 and 0.1197 in native normalized controller units. Gemini's direct seven-value response has MAE 0.2411; its gripper sign error is 0.3169, compared with 0.0949 for B2. Gemini semantics plus local grounding have MAE 0.1323. State-only and shuffled-semantic controls have 0.1372 and 0.1333. The 0.0010 real-versus-shuffled difference is small; no paired episode-level interval is available in the saved summary, so we make no content-specific gain claim. The hybrid MAE is 0.1308. TypeSafe Jev has no action MAE because it never produced a seven-value action. Its E3b role was selection.

E3's semantic probes improve on majority direction baselines for y (0.691 versus 0.556) and z (0.917 versus 0.793), but not x (0.590 versus 0.657). Gripper-event probe accuracy equals the no-change baseline (both 0.956). Thus the probes show some coarse signal, not dependable contact understanding. The two-frame Gemini subset was collected but not evaluated as a trained downstream policy ablation.

On the 527 E3b held-out transitions, the hybrid action's bad-event rate is 0.670. Hosted Jev's calibrated error-risk ranking has AUC 0.491 and Brier score 0.239; the matched Gemini-confidence

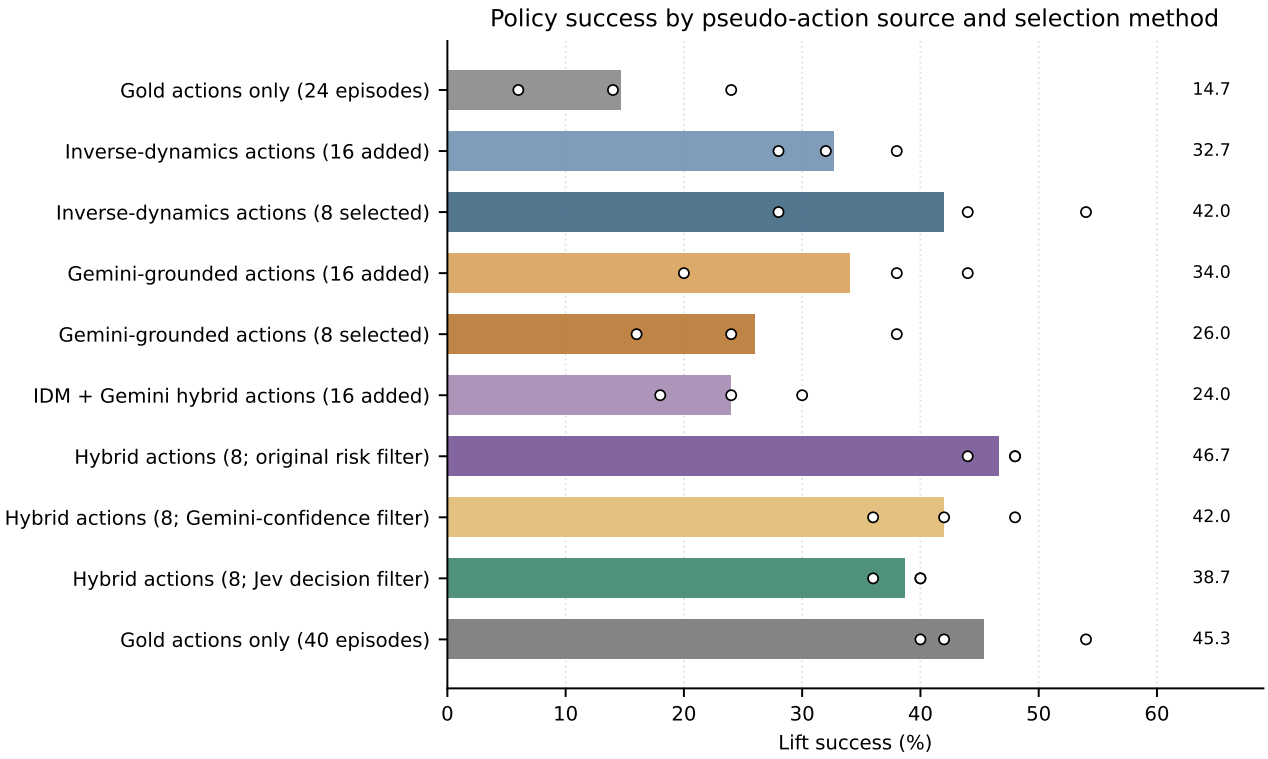


Figure 1: All policy arms. Every pseudo-action row adds the indicated episodes to the same 24 gold episodes. The Jev row uses Jev only to select episodes; its actions come from the saved inverse-dynamics/Gemini hybrid. White points show the three training-seed success rates.

scores have 0.508 and 0.233; the existing hybrid score has 0.680 and 0.214. The constant calibration base-rate Brier is 0.233. Jev's Noul values are probabilities about the worded ambiguity propositions, *not* probabilities of motor error; the isotonic map attempts that translation. Near-0.5 AUC means this chosen hosted decision task provided no useful held-out error ranking here. The E2 B2 risk map's held-out AUC was 0.695. Figure 4 shows the E3b selection curves; scoring individual transitions for this diagnostic differs from ranking whole episodes for policy training.

5 Interpretation and limitations

These experiments show that pseudo-action data can be useful in this Lift setting; they do not isolate a necessary contribution from pretrained semantics. Filtering can change training-set difficulty, number of examples, and action-error distribution at once. Therefore a filtered policy's gain cannot be attributed solely to better uncertainty calibration or to the visual model. The historical filtering implementation has a tied-score pooling defect in its isotonic helper. E3b used a corrected implementation for *both new selectors*; historical E3 results were preserved rather than silently recomputed. Consequently hosted Jev versus Gemini confidence is the cleaner comparison of new selectors.

The hosted study is particularly narrow. Jev read Gemini's already compressed semantic JSON; it had no access to pixel evidence omitted or misread by Gemini. Repeated descriptions collapse 1,646 transitions into 72 distinct hosted requests, restricting input diversity. Neither a result for typed text decisions nor a result for one 24-label robot task measures general hosted-Jev decision ability, direct visual control, cross-embodiment transfer, or world-model prediction. E3b reused a pilot evaluation popula-

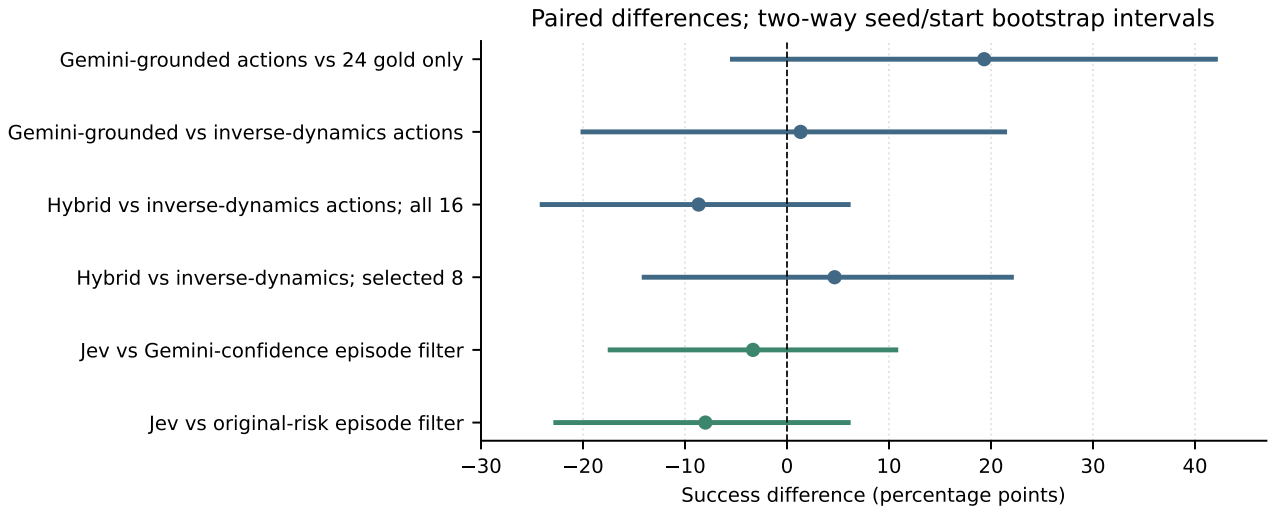


Figure 2: Selected paired success differences with two-way bootstrap intervals resampling three policy seeds and 50 shared starts (10,000 draws, fixed RNG seed 731). The first three contrasts were E3 primary comparisons; the matched-filter E3 contrast is exploratory. The last two compare episode selectors applied to identical saved hybrid actions, not different action generators. Intervals describe this pilot and are not equivalence tests.

tion observed during E3. The 40-episode test remained locked; three BC seeds and one visual/grounding seed limit generalization. No methodological or statistical equivalence margin was prespecified.

The Gemini production pass also required documented integrity repairs. After 539 legacy semantic cache entries, scene-specific episode XML was restored before rendering; the old entries were excluded and the corrected cache was rebuilt. Later action-alignment and CPU/MPS device fixes occurred before the affected downstream results. These were repairs rather than prespecified model improvements. The final E3 VLM summary counted only the last resume pass (196 requests), not total campaign usage. Across cache ledgers, 2,751 semantic, 1,646 direct, and 190 two-frame successful annotations are covered. Cache tokens imply approximately $26.23 at listed Gemini prices; a read-only key query immediately after E3 showed $26.61 cumulative key usage including earlier invalidated calls, smoke tests, and retries. E3b's TypeSafe decisions added $0.00196014, with no repeat Gemini calls. Billing uncertainty prevents assigning every cent of the cumulative key amount to the final valid dataset.

6 Related work and conclusion

Latent action pretraining from unlabeled video followed by action adaptation has established precedents: LAPA learns quantized latent actions before VLA pretraining and robot-action fine-tuning (Ye et al., 2025); UniVLA learns task-centric latents in DINO feature space with language context (Bu et al., 2025); and latent-action world models in the wild use constrained continuous representations and action adapters (Garrido et al., 2026). CD-LAM studies how irrelevant visual changes can contaminate learned action latents (Wei et al., 2026). Our single-task, small-budget pseudo-labeling and BC study does not reproduce those systems, support a first-method claim, or make cross-paper leaderboard comparisons.

The evidence supports a bounded conclusion: local inverse-dynamics pseudo-labels improved Lift policy success over 24 gold demos. Gemini semantics plus small grounding also improved the point estimate over 24 gold demos, but did not establish an advantage over B2. A later hosted TypeSafe Jev decision layer did not improve selection of the saved hybrid pseudo-actions over the matched Gemini-

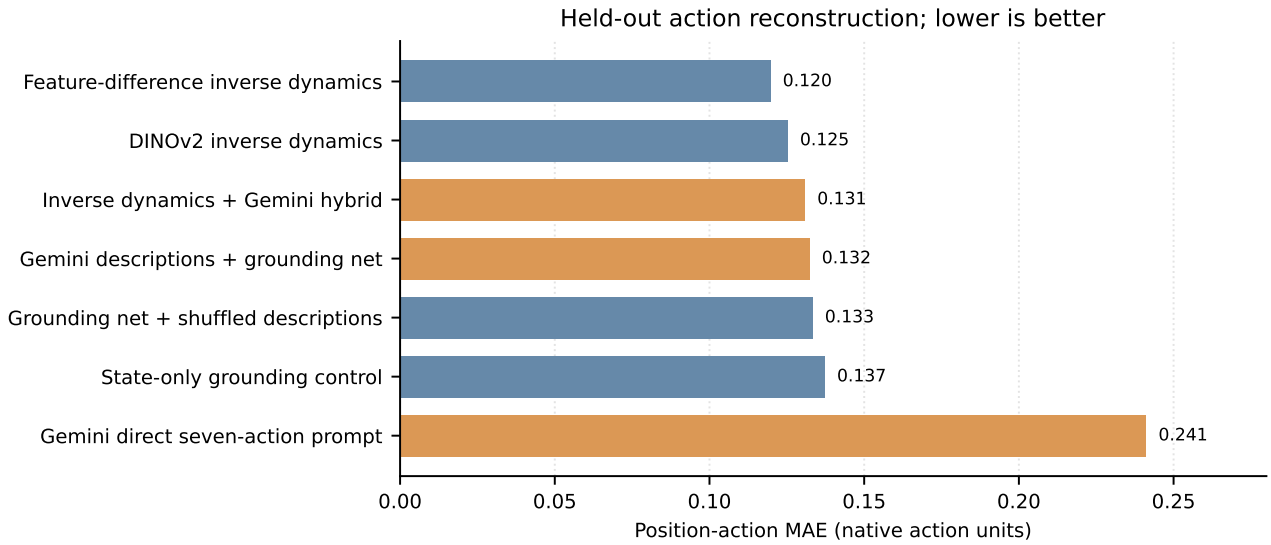


Figure 3: Action reconstruction on the 12-episode evaluation holdout. Values are MAE in native action units, not percentages. These diagnostic scores need not order policies by closed-loop success.

confidence control in this pilot. The next decisive study should freeze a new contact-sensitive selector, replicate across independent model-training seeds and a second task, and evaluate on previously unseen episodes. Its advancement criterion should require better calibrated held-out error ranking *and* paired policy utility above a matched local filtered baseline; failure on either rejects that selector's claimed data-utility gain. A larger world-model claim requires an actual action-conditioned world-model consumer.

A Reproducibility and provenance

The frozen configuration and implementation are in `config/`, `jev_research/`, and `research/jev/`. Lightweight run JSON and paired rollout records used in this paper are committed under `results/runs/`. Figure code is `figs/make_policy_fig.py`; it reads those JSON files without model inference. The E3b client and runner are `research/jev/hosted_jev.py` and `research/jev/run_hosted_jev.py`. Training/test access rules are implemented in `jev_research/data.py` and `jev_research/policy.py`. The dataset/controller audit is `results/runs/lift_e0_none_seed0/e0_report.json`. E3b has 383 calibration and 527 held-out transitions for the hybrid bad-event diagnostic. The two-way intervals in Fig. 2 resample three policy-seed indices and 50 shared start indices, with 10,000 draws and RNG seed 731. This sensitivity analysis is descriptive, since three trained policies cannot precisely identify between-seed variance.

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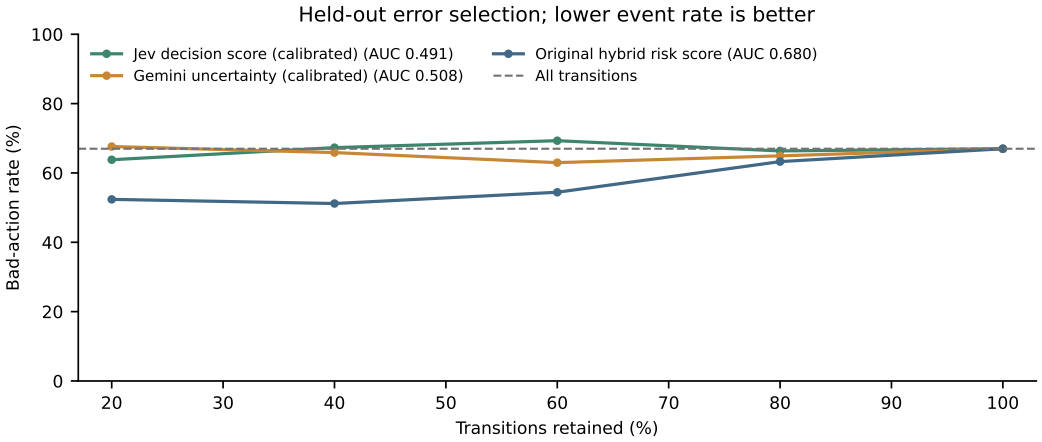


Figure 4: Bad-action rate as the lowest-risk fraction of held-out transitions is retained. All three plotted scores refer to the same saved hybrid action predictions. The old hybrid map is unchanged; the two new selectors use the same tied-score-correct isotonic fit.

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